

LABORATORY ACTIVITY 6

Data Portability, Automated Operations, and Advanced System Architecture

Course: Web Development with Laravel

Project Title: PageTurner Online Bookstore Management System

Laboratory Type: Hands-on / Project-Based

Difficulty Level: Advanced to Expert

Prerequisite: Completed Laboratory Activity 4

1. Laboratory Background

In Laboratory Activity 4, students implemented enterprise-grade authentication, security hardening, email workflows, and role-based dashboards. The system now has a robust foundation with 2FA, email verification, and comprehensive notification systems.

In **Laboratory Activity 6**, the system is extended to include **enterprise data management capabilities** critical for production e-commerce platforms: bulk data import/export operations, automated database backup scheduling, comprehensive audit logging, API rate limiting, and advanced data transformation features. This laboratory simulates real-world requirements for scalable, maintainable, and compliant business applications

.

2. Learning Objectives

At the end of this laboratory activity, students will be able to:

1. **Implement robust import/export systems** using Laravel Excel with chunked processing, validation, and queuing for large datasets
 2. **Design automated backup strategies** using Laravel's task scheduler and Spatie Backup package with monitoring and notifications
 3. **Apply comprehensive audit logging** to track sensitive data changes and maintain compliance
 4. **Implement API rate limiting** with per-second granularity and user-based tiered limits
 5. **Configure database read/write splitting** for performance optimization
 6. **Build data transformation pipelines** using middleware for API request/response handling
 7. **Create scheduled maintenance tasks** for data cleanup, report generation, and system health checks
-

3. System Scope and Requirements

3.1 Existing System Context (From Laboratory Activity 4)

The PageTurner system currently supports:

- Three user types: Administrators, Registered Customers, General Visitors
- Advanced authentication: Email verification, 2FA, password recovery
- Role-based dashboards with real-time statistics
- Email notification system for orders, reviews, and security events
- Policy-based authorization and middleware protection

3.2 New Functional Requirements

4. Data Import/Export System (4.1)

4.1.1 Book Import/Export Module

Import Features:

- **Bulk book uploads** via Excel/CSV (XLSX, CSV formats)
- **Template validation** with required headers: ISBN, Title, Author, Price, Stock, Category, Description
- **Data validation rules:**
 - ISBN: Unique, valid format (ISBN-10 or ISBN-13)
 - Title: Required, max 255 characters
 - Price: Numeric, positive, max 9999.99
 - Stock: Integer, non-negative
 - Category: Must exist in categories table
- **Chunked processing** for files > 1000 rows (chunk size: 1000)
- **Queue integration** for background processing with progress tracking
- **Error handling:** Skip-on-failure with detailed failure reports
- **Duplicate detection:** Option to update existing books or skip duplicates

Export Features:

- **Filtered exports:** By category, price range, stock status, date range
- **Format options:** XLSX, CSV, PDF
- **Chunked exports** for large datasets using FromQuery concern
- **Queued exports** for datasets > 10,000 records
- **Custom column selection:** Allow users to choose which fields to export

Technical Requirements:

- Use maatwebsite/excel package (Laravel Excel)
- Implement ShouldQueue for large files
- Apply WithBatchInserts (batch size: 1000) for database optimization
- Create downloadable import templates
- Store import/export logs in database

4.1.2 Order Export Module

- **Admin order exports:** Filter by status, date range, customer
- **Customer order history export:** Personal orders only (PDF invoice generation)

- **Financial reporting exports:** Revenue summaries, tax reports
- **Scheduled exports:** Daily sales reports emailed to administrators

4.1.3 User Import/Export (Admin Only)

- **Bulk user creation** for corporate/institutional accounts
- **User data export** with PII redaction options for GDPR compliance
- **Role assignment** during import

5. Automated Backup and Maintenance System (4.2)

4.2.1 Database Backup Scheduling

Backup Configuration:

- **Automated daily backups** at 02:00 AM (avoiding DST issues)
- **Weekly full backups** every Sunday
- **Retention policy:** Keep 7 daily backups, 4 weekly backups, 12 monthly backups
- **Storage destinations:** Local (encrypted), S3-compatible cloud storage
- **Backup contents:** Database dump, uploaded files (book covers), system configuration

Implementation Requirements:

- Use spatie/laravel-backup package
- Configure backup monitoring with health checks
- **Failure notifications:** Email alerts on backup failure
- **Success notifications:** Weekly backup summary reports
- **Manual backup trigger:** Admin dashboard button for immediate backup

4.2.2 Maintenance Scheduling

Scheduled Tasks (Laravel Scheduler):

Table

Task	Frequency	Description
backup:run	Daily at 02:00	Full database and files backup
backup:clean	Daily at 03:00	Remove old backups per retention policy
order:cleanup-pending	Hourly	Cancel pending orders > 24 hours old
session:cleanup	Daily	Clear expired sessions
log:rotate	Weekly	Archive and compress old logs
report:generate-daily	Daily at 06:00	Generate daily sales report
notification:prune	Weekly	Delete old notification records > 90 days
audit:archive	Monthly	Archive audit logs > 1 year old

Technical Requirements:

- Use withoutOverlapping() for long-running tasks
- Implement task hooks for onFailure() and onSuccess()
- Log all scheduled task execution results

- Configure server cron job: `* * * * * cd /path && php artisan schedule:run >> /dev/null 2>&1`

6. Audit Logging and Compliance (4.3)

4.3.1 Comprehensive Audit System

Audited Events:

- **Authentication events:** Login, logout, failed attempts, password changes, 2FA enable/disable
- **Data modifications:** Book CRUD, category changes, order status updates, review moderation
- **Security events:** Permission changes, role assignments, email verification
- **System events:** Backup operations, imports/exports, settings changes

Audit Log Structure:

php

Copy

```
{
  "id": "uuid",
  "user_id": 123,
  "event": "updated",
  "auditable_type": "App\\Models\\Book",
  "auditable_id": 456,
  "old_values": {"price": 19.99, "stock": 50},
  "new_values": {"price": 24.99, "stock": 45},
  "metadata": {
    "ip_address": "192.168.1.100",
    "user_agent": "Mozilla/5.0...",
    "url": "/admin/books/456",
    "method": "PUT"
  },
  "created_at": "2026-04-13T14:30:00Z"
}
```

Implementation Requirements:

- Use owen-it/laravel-auditing or iamfarhad/laravel-audit-log package
- **Global scoping:** Exclude sensitive fields (passwords, tokens, payment info) by default
- **Custom audit events:** For complex business logic (order status transitions)
- **Retention policy:** 1 year online, 5 years archived storage
- **Tamper-proof storage:** Write-once audit log table with checksums

4.3.2 Audit Log Dashboard (Admin Only)

- **Searchable/filterable logs:** By user, date range, event type, affected model
- **Diff visualization:** Side-by-side comparison of old/new values
- **Export capability:** CSV/PDF export of audit trails for compliance
- **Real-time alerts:** Email notifications for critical security events (permission changes, admin actions)

7. API Rate Limiting and Security (4.4)

7.1 Tiered Rate Limiting Strategy

Rate Limit Tiers:

Table

Tier	Requests/Minute	Scope	Users
public	30	General browsing, book search	Visitors
standard	60	Authenticated API access	Regular customers
premium	300	High-volume API access	Premium/VIP customers
admin	1000	Administrative operations	Administrators
auth	10	Login, registration, password reset	All (strict)

Implementation Requirements:

- **Per-second granularity** for burst protection
- **User-based limiting:** By user ID for authenticated, IP for guests
- **Dynamic limits:** Check user role/subscription tier in RateLimiter
- **Response headers:** X-RateLimit-Limit, X-RateLimit-Remaining, Retry-After
- **Custom error responses:** JSON 429 responses with limit information

7.2 API Resource Optimization

Request/Response Transformation:

- **Middleware-based transformation:** Convert snake_case (database) to camelCase (JSON) automatically
- **Field filtering:** Allow clients to specify required fields (?fields=id,title,price)
- **Pagination:** Cursor-based pagination for large collections
- **Caching:** ETag support for conditional requests

8. Database Architecture Enhancements (4.5)

8.1 Read/Write Splitting (Optional Advanced Feature)

Configuration:

- **Write master:** Primary database for all writes
- **Read replicas:** 1-2 read replicas for query load distribution
- **Sticky sessions:** Enable sticky option for immediate read-after-write consistency

Implementation:

php

Copy

```
'mysql' => [  
  'read' => [  
    'host' => ['192.168.1.2', '192.168.1.3'],  
  ],  
  'write' => [  
    'host' => ['192.168.1.1'],  
  ],  
  'sticky' => true,  
  // ... other config  
,
```

8.2 Connection Pooling and Optimization

- **Database indexing strategy:** Add indexes for frequent query patterns (ISBN, category_id, user_id, order_date)
- **Query caching:** Cache frequent read queries (book categories, bestseller lists)
- **Lazy loading prevention:** Eager load relationships in export queries

9. Advanced Dashboard Enhancements (4.6)

9.1 Admin Dashboard - Data Management Section

New Widgets:

- **Import/Export Status:** Recent operations, queue status, success/failure rates
- **Backup Status:** Last backup time, size, storage location, health indicator
- **Audit Log Summary:** Recent critical events, security alerts
- **API Usage Statistics:** Requests per endpoint, rate limit hits, error rates
- **System Health:** Database size, storage usage, queue lengths, failed job counts

9.2 User Dashboard - Data Portability

New Features:

- **Export my data:** GDPR-compliant personal data export (JSON format)
- **Order history export:** PDF/Excel download of all orders
- **Reading history:** Export book browsing/purchase history

10. Database Enhancements

10.1 New/Modified Tables

Table

Table	Purpose
import_logs	Track import operations: filename, rows processed, failures, user_id
export_logs	Track export requests: format, filters, status, download link
audit_logs	Comprehensive change tracking (if using custom implementation)

scheduled_tasks	Custom schedule monitoring beyond Laravel's default
api_rate_limits	Track rate limit hits and throttling events
backup_monitoring	Backup execution logs, verification status

10.2 Migration Requirements

- All changes must use Laravel migrations with rollback support
- Seeders for test data (sample import files, mock audit logs)
- Foreign key constraints for data integrity

11. Testing and Validation

Students must demonstrate:

11.1 Import/Export Testing

- ☒ Successful import of 10,000+ book records with validation
- ☒ Handling of malformed files with proper error reporting
- ☒ Queue processing and background job completion
- ☒ Memory usage stays < 256MB for large imports (chunking verification)
- ☒ Export of 50,000+ records without timeout (streaming/queuing)

11.2 Backup and Scheduling Testing

- ☒ Manual backup execution and verification
- ☒ Scheduled task automation (using php artisan schedule:run)
- ☒ Backup file integrity and restoration procedure
- ☒ Failure notification delivery
- ☒ Retention policy enforcement

11.3 Audit and Compliance Testing

- ☒ All CRUD operations create appropriate audit entries
- ☒ Sensitive data exclusion from logs (passwords redacted)
- ☒ Audit log search and filtering functionality
- ☒ Tamper-proof verification (checksums)

11.4 Rate Limiting Testing

- ☒ Tiered limits enforced correctly per user role
- ☒ 429 responses with proper headers
- ☒ Graceful degradation under load
- ☒ Per-second burst protection

12. Deliverables

1. **Complete source code repository** with:
 - Import/Export classes using Laravel Excel concerns

- Custom Artisan commands for scheduled tasks
 - Rate limiter configurations
 - Middleware for data transformation
2. **Database migrations and seeders** for all new tables
 3. **Configuration files:**
 - config/backup.php - Backup scheduling and storage
 - config/excel.php - Import/export settings
 - config/audit.php - Audit logging configuration
 4. **Documentation:**
 - **Technical report** (1500+ words) covering:
 - Architecture decisions for chunking and queuing strategies
 - Backup strategy and disaster recovery procedures
 - Security considerations for audit logging
 - Performance optimization techniques
 - **API documentation** for rate limits and data endpoints
 - **User guide** for import/export operations
 5. **Screenshots/recordings showing:**
 - Bulk book import with progress tracking
 - Failed import error handling
 - Automated backup configuration
 - Audit log dashboard
 - Rate limiting in action (429 responses)

13. Grading Rubric

Table

Component	Percentage	Criteria
Import/Export System	25%	Functionality, validation, chunked processing, queue integration, error handling
Backup & Automation	20%	Scheduling configuration, monitoring, notifications, retention policies
Audit & Compliance	20%	Comprehensive logging, sensitive data handling, dashboard, tamper-proofing
API Rate Limiting	15%	Tiered limits, per-second granularity, proper responses, user-based differentiation
Advanced Features	10%	Read/write splitting, data transformation, performance optimization

Documentation & Testing	10%	Technical report quality, test coverage, code organization
------------------------------------	-----	--

14. Expected Learning Outcomes

Upon completion of Laboratory Activity 6, students will have built a **production-grade data management layer** featuring:

- **Enterprise data portability** with robust import/export capabilities handling millions of records
- **Disaster recovery preparedness** through automated backup strategies and monitoring
- **Compliance readiness** with comprehensive audit trails for regulatory requirements (GDPR, PCI DSS)
- **API resilience** through sophisticated rate limiting preventing abuse and ensuring fair usage
- **Performance optimization** via database architecture improvements and efficient data processing

This laboratory prepares students for **senior backend engineering roles** requiring expertise in data-intensive applications, system reliability, and enterprise security compliance.

15. Suggested Packages and Tools

Table

Purpose	Package	Version
Excel Import/Export	maatwebsite/excel	^3.1
Database Backup	spatie/laravel-backup	^9.0
Audit Logging	owen-it/laravel-auditing	^13.0
Rate Limiting	Built-in Laravel 11	-
Task Scheduling	Built-in Laravel Scheduler	-
PDF Generation	barryvdh/laravel-dompdf	^2.0
Queue Monitoring	laravel/horizon	^5.0